随机过程

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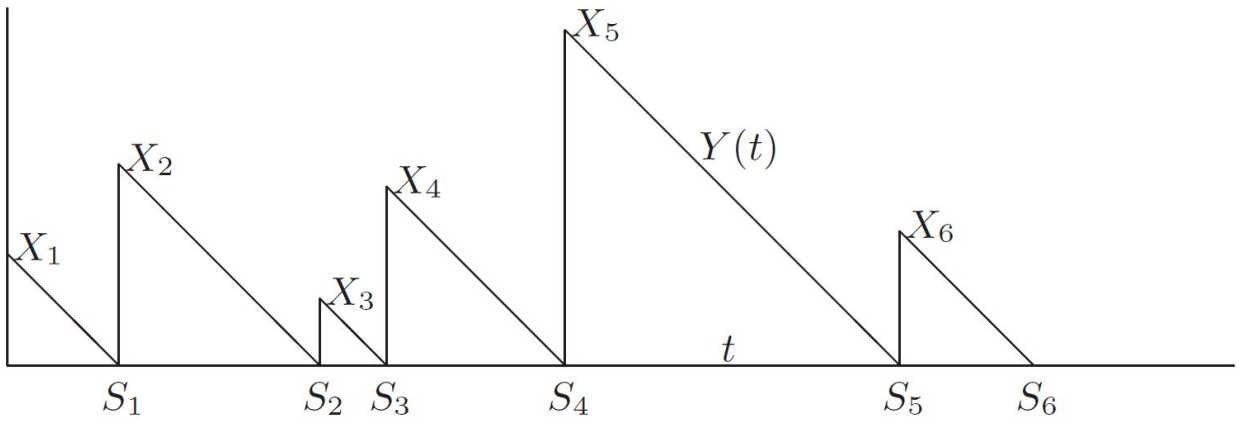
Review

§ 5.5 Stopping times for repeated experiments

Review

Definition 1

For a renewal process $\{N(t); t > 0\}$, the residual life $Y(t)$ at time t is the remaining time until the next renewal, i.e., $Y(t) = S_{N(t)+1} - t$.

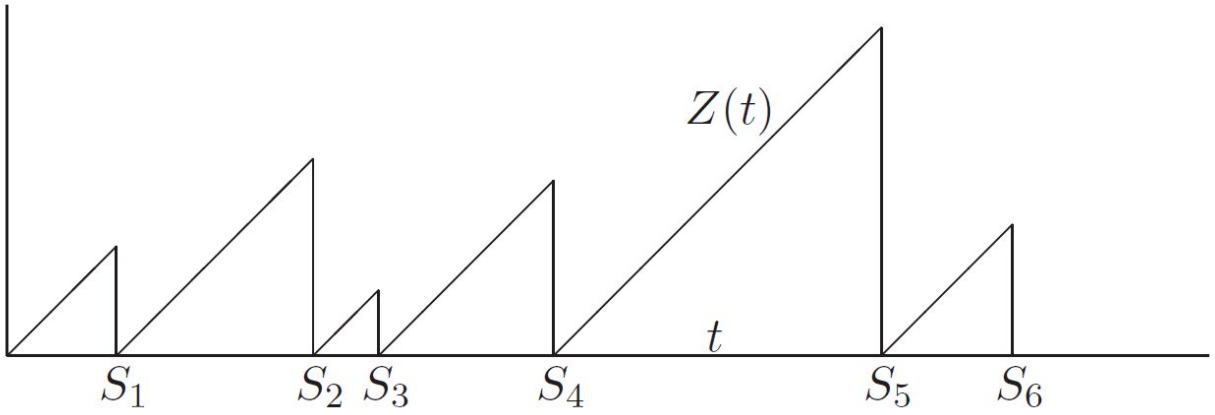


$$\lim_{t\to\infty} \frac{\int_0^t Y(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

Review

Definition 2

Let $Z(t)$ be the age of a renewal process at time t where age is defined as the interval from the most recent arrival before (or at) t until t, i.e., $Z(t) = t - S_{N(t)}$.



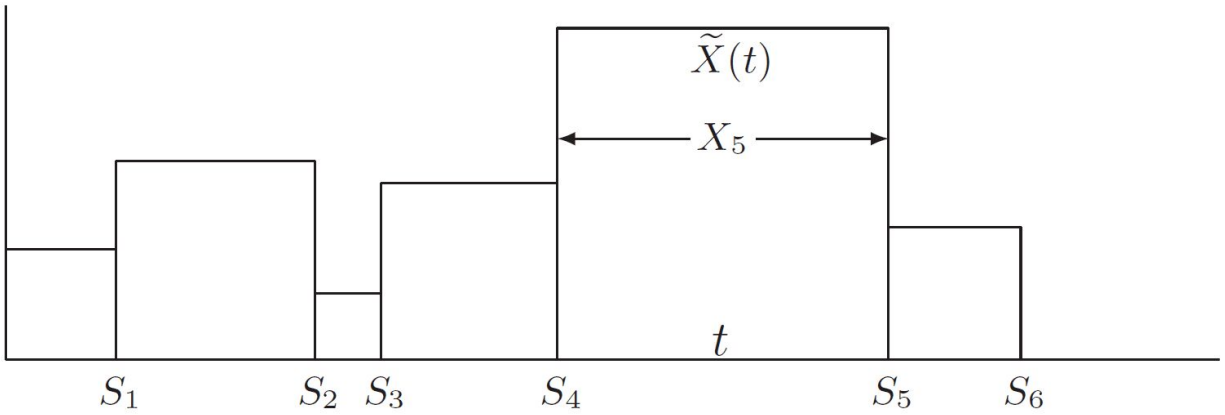
$$\lim_{t\to\infty} \frac{\int_0^t Z(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{2\mathrm{E}[X]} \qquad \text{WP1.}$$

Review

Definition 3

Let $\widetilde{X}(t)$ be the duration of the inter-renewal interval containing time t, i.e., $\widetilde{X}(t) = X_{N(t)+1} = S_{N(t)+1} - S_{N(t)}$.

It is clear that $\widetilde{X}(t) = Y(t) + Z(t)$.



$$\lim_{t\to\infty} \frac{\int_0^t \widetilde{X}(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[X^2]}{\mathrm{E}[X]} \qquad \text{WP1.}$$

Review

Consider the reward $R(t)$ at time t is given explicitly as a function $R(Z(t), \widetilde{X}(t))$ of the age and duration at t.

We first define R_n as the accumulated reward in the nth renewal interval,

$$R_n \triangleq \int_{S_{n-1}}^{S_n} R(\tau)\mathrm{d}\tau = \int_{z=0}^{X_n} R(z, X_n)\mathrm{d}z.$$

This is a function only of the rv. X_n.

Theorem 1

Let $\{R(t); t > 0\}$ be a renewal-reward function for a renewal process with expected inter-renewal time $\mathrm{E}[X] = \overline{X} < \infty$. If each R_n is a r.v. with $\mathrm{E}[|R_n|] < \infty$, then

$$\lim_{t\to\infty} \frac{1}{t}\int_{\tau=0}^{t} R(\tau)\mathrm{d}\tau = \frac{\mathrm{E}[R_n]}{\overline{X}} \quad \text{WP1.}$$

Chapter 5. Renewal processes

Review

§ 5.5 Stopping times for repeated experiments

5.5.1 Wald's equality

5.5.2 Applying Wald's equality to $\mathrm{E}[N(t)]$

5.5.3 Generalized stopping trials, embedded renewals and G/G/1 queues

5.5.4 Little's theorem

5.5.5 M/G/1 queues

§ 5.5 Stopping times for repeated experiments

Definition 4

A stopping trial (or stopping time) J for a sequence $\{X_n; n \geqslant 1\}$ of r.v.s is a positive integer-valued r.v. such that for each $n \geqslant 1$, the indicator r.v. $\mathbb{I}_{\{J=n\}}$ is a function of $\{X_1, X_2, \ldots, X_n\}$. (We can relax the definition further to allow J to be a possibly defective r.v..)

§ 5.5 Stopping times for repeated experiments

Examples:

- A gambler goes to a casino and gambles until broke.
- Flip a coin until 10 successive heads appear.
- Test an hypothesis with repeated trials until one or the other hypothesis is sufficiently probable a posteriori.
- 我们可能需要从几个假设中选择一个，为此需要重复做试验直到足以区分这些假设。如果允许这个试验次数依赖于试验结果，则试验次数通常只是预先设定的次数的很小一部分。
- 树搜索过程中，如果某些路径再搜索下去没有收益的话，也可以停下来。

§ 5.5 Stopping times for repeated experiments

独立掷一个均匀的骰子。令 X_n 表示第 n 次所得点数。令 $S_n = X_1 + X_2 + \cdots + X_n$。定义一个正整数变量 J，$J = n$ 当且仅当 $S_n > 6$.

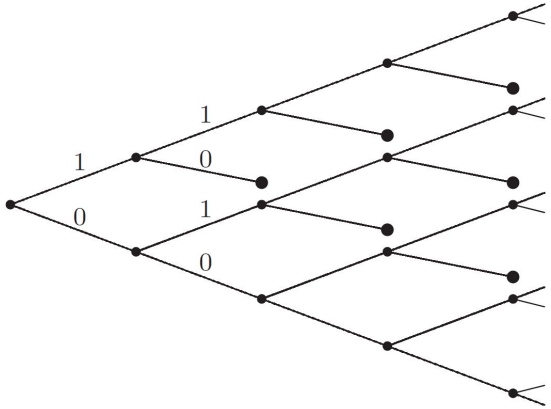
- 说明 J 是一个停止时间。
- 计算 $\{J < 3\}$ 的概率。
- 计算 $\{J \geq 3\}$ 的概率。
- 计算在给定 $X_3 = 6$ 的条件下，$\{J \geq 3\}$ 的概率。

§ 5.5 Stopping times for repeated experiments

Suppose the rv.s X_i in a process $\{X_n; n \geqslant 1\}$ have a finite number of possible sample values. Then any (possibly defective) stopping trial J can be represented as a rooted tree where the trial at which each sample path stops is represented by a terminal node.

Exmaple 3

For a Bernoulli process $\{X_n; n \geqslant 1\}$, stopping occurs when the pattern $(1, 0)$ first occurs.



5.5.1 Wald's Equality

Theorem 2 (Wald's equality)

Let $\{X_n; n \geqslant 1\}$ be a sequence of i.i.d. r.v.s, each of mean $\overline{X}$.

If J is a stopping trial for $\{X_n; n \geqslant 1\}$ and if $\mathrm{E}[J] < \infty$, then the sum

$S_J = X_1 + X_2 + \cdots + X_J$ at the stopping trial J satisfies

$$\mathrm{E}[S_J] = \overline{X}\mathrm{E}[J].$$

思考题:

掷一个骰子，点数记为 J。接着再掷 J 次骰子，点数分别记为

$X_1, X_2, \cdots, X_J$。若骰子是均匀的，且各次投掷是独立的，求 $\mathrm{E}[S_J]$。

5.5.1 Wald's Equality

Theorem 2 (Wald's equality)

Let $\{X_n; n \geqslant 1\}$ be a sequence of i.i.d. r.v.s, each of mean $\overline{X}$.
If J is a stopping trial for $\{X_n; n \geqslant 1\}$ and if $\mathrm{E}[J] < \infty$, then the sum $S_J = X_1 + X_2 + \cdots + X_J$ at the stopping trial J satisfies

$$\mathrm{E}[S_J] = \overline{X}\mathrm{E}[J].$$

Proof:

$$S_J = \sum_{n=1}^{J} X_n = \sum_{n=1}^{\infty} X_n \mathbb{I}_{\{J \geqslant n\}}.$$

$$\mathrm{E}[S_J] = \mathrm{E}\left[\sum\nolimits_n X_n \mathbb{I}_{\{J \geqslant n\}}\right] = \sum\nolimits_n \mathrm{E}\left[X_n \mathbb{I}_{\{J \geqslant n\}}\right]$$

The essence of the proof is to show that X_n and $\mathbb{I}_{\{J \geqslant n\}}$ are independent.

5.5.1 Wald's Equality

To show that X_n and $\mathbb{I}_{\{J \geqslant n\}}$ are independent, note that

$$\mathbb{I}_{\{J \geqslant n\}} = 1 - \mathbb{I}_{\{J < n\}}.$$

Also $\mathbb{I}_{\{J<n\}}$ is a function of $X_1, \ldots, X_{n-1}$. Since the is X_i are i.i.d., X_n is independent of $X_1, \ldots, X_{n-1}$, and thus $\mathbb{I}_{\{J<n\}}$, and thus of $\mathbb{I}_{\{J \geqslant n\}}$. Therefore,

$$\begin{aligned}
\mathrm{E}[S_J] &= \sum_n \mathrm{E}[X_n \mathbb{I}_{\{J \geqslant n\}}] \\
&= \sum_n \mathrm{E}[X_n]\mathrm{E}[\mathbb{I}_{\{J \geqslant n\}}] \\
&= \overline{X} \sum_n \mathrm{E}[\mathbb{I}_{\{J \geqslant n\}}] \\
&= \overline{X} \sum_n \Pr\{J \geqslant n\} = \overline{X}\mathrm{E}[J].
\end{aligned}$$

5.5.1 Wald's Equality

思考题：成功概率 p 的 Bernoulli 试验，首次成功平均需要几次试验？你能想到多少种解法？

Example 4: Stop when you are ahead

Consider tossing a coin with probability of heads equal to p. \$1 is bet on each toss and you win on heads, lose on tails. You stop when your winnings reach \$1.

We can model it as a sequence of i.i.d. rv.s $X_1, X_2, \ldots$, where each X_i is 1 with probablity p and -1 with probablity $1-p$. Consider the possibly-defective stopping trial J, where J is the first n for which $S_n = X_1 + \cdots + X_n = 1$, i.e., the first trial at which the gambler is ahead.

5.5.1 Wald's Equality

令 $\theta = \Pr\{J < \infty\}$. 则 $\theta = p + (1-p)\theta^2$, 解得 $\theta = 1$ 或者 $\theta = p/(1-p)$.
Consider $p > 1/2$.
对应的解是 $\theta = 1$, 因为后者不是概率。从另外角度，Your winnings (in the absence of stopping) would grow without bound, passing through 1, so J must be a rv.. $S_J = 1$ WP1, so $\mathrm{E}[S_J] = 1$.
Thus, we have

$$\mathrm{E}[J] = 1/\overline{X} = \frac{1}{2p-1}.$$

On the other hand, note that $J = 1$ with probability p. If $J > 1$, i.e., if $S_1 = -1$, then the only way to reach $S_n = 1$ is to go from $S_1 = -1$ to $S_m = 0$ for some m (requiring $\overline{J}$ steps on average); $\overline{J}$ more steps on average then gets to 1. Thus

$$\overline{J} = p + (1-p)(1 + 2\overline{J}),$$

which implies $\overline{J} = \frac{1}{2p-1}$.

5.5.1 Wald's Equality

Consider $p < 1/2$.

$\theta = p/(1-p)$ 是正确的解, $\theta = 1$ 不是。Thus J is defective and Wald's equation is inapplicable.

5.5.1 Wald's Equality

Consider $p = 1/2$.

In the limit as p approaches $1/2$ from below,

$\Pr\{J < \infty\} = p/(1-p) \to 1$. However, as p approaches $1/2$ from above,

we see that $\mathrm{E}[J] = 1/(2p-1) \to \infty$.

Wald's equality does not hold here, since $\mathrm{E}[J] = \infty$, and in fact does not make sense since $\overline{X} = 0$.

领先的概率是 1，但有两个问题：一，平均时间是无穷；二、潜在资本风险。

5.5.3 Generalized stopping trials

Definition 5

A generalized stopping trial J for a sequence of pairs of r.v.s $(X_1, Y_1), (X_2, Y_2), \ldots$ is a positive integer-valued r.v. such that, for each $n \geqslant 1$, $\mathbb{I}_{J=n}$ is a function of $X_1,\ Y_1,\ X_2,\ Y_2, \ldots, X_n,\ Y_n$.

It follows that $\mathbb{I}_{J<n} = 1 - \mathbb{I}_{J\geqslant n}$ is a function of $X_1,\ Y_1, \ldots, X_{n-1},\ Y_{n-1}$.

Theorem 3 (Generalized Wald's equality)

Let $\{(X_n, Y_n); n \geqslant 1\}$ be a sequence of pairs of r.v.s, where each pair is i.i.d. to all other pairs. Assume that each X_i has finite mean $\overline{X}$. If J is a stopping trial for $\{(X_n, Y_n); n \geqslant 1\}$ and if $\mathrm{E}[J] < \infty$, then the sum $S_J = X_1 + X_2 + \cdots + X_J$ satisfies

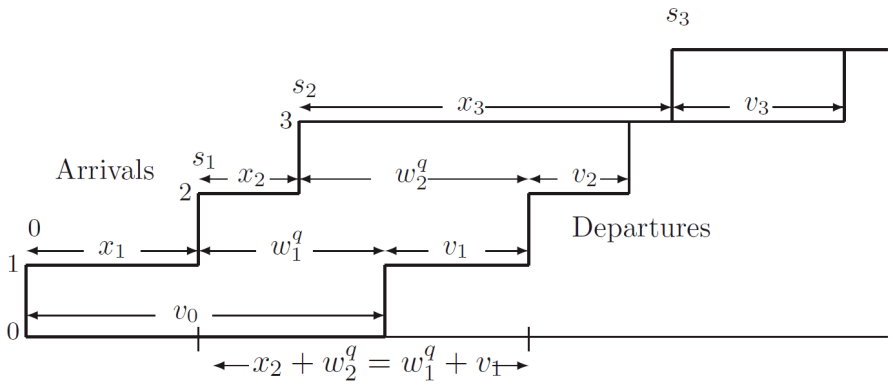
$$\mathrm{E}[S_J] = \overline{X}\mathrm{E}[J].$$

Remark: Also each Y_i can be replaced by a vector of rv.s.

5.5.3 G/G/1 queues

- The interarrival times are i.i.d.. Let $\{X_i; i \geqslant 1\}$ denote the sequence of i.i.d. interarrival intervals, where X_i is the interval from the arrival of customer $i-1$ to i.
- The service times are i.i.d. and independent of the interarrival times. Let $\{V_i; i \geqslant 0\}$ denote the i.i.d. required service times for these customers; these service times are independent of the interarrival times.
- Customer 0 arrives at time 0 and immediately enters the server. Subsequent arrivals are stored in a queue if the server is busy.
- Each enqueued arrival enters the server, in first-come-first-served (FCFS) order, when the server completes the previous service.

5.5.3 G/G/1 queues



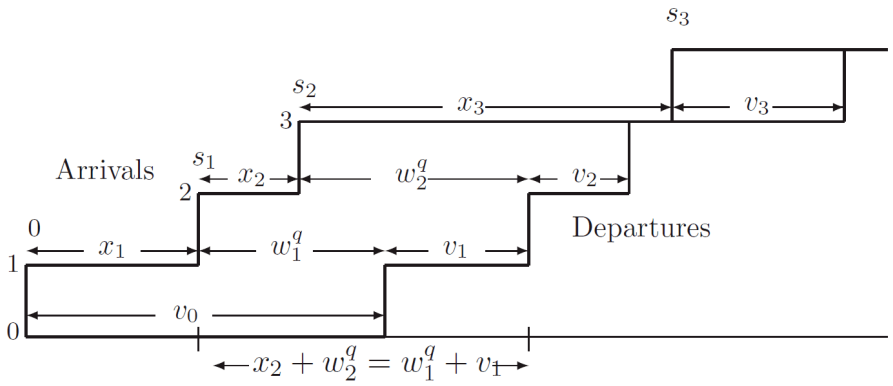
- For Customer 1, if $X_1 < V_0$, then the first arrival waits in queue for $W_1^q = V_0 - X_1$. Otherwise, customer 1 enters service immediately. Thus

$$W_1^q = \max\{V_0 - X_1, 0\}.$$

- For customer i, （在系统里呆的时间 = 排队时间 + 窗口时间）

$$W_i^q = \max\{W_{i-1}^q + V_{i-1} - X_i, 0\}.$$

5.5.3 G/G/1 queues



Consider the first arrival (s_3 above) that starts a new busy period as a generalized stopping trial.

- The sequence of paired rv. is $(X_1, V_0), (X_2, V_1), \ldots$.
- J is the smallest n for which $\sum_{i=1}^{n}(X_i - V_{i-1}) \geqslant 0$.
- If $\mathrm{E}[X] > \mathrm{E}[V]$, then $\lim_{n\to\infty} \Pr\{\sum_{i=1}^{n}(X_i - V_{i-1}) \geqslant 0\} = 1$. Thus J is a non-degenerate stopping trial.
- If $\mathrm{E}[J] < \infty$, then the generalized Wald's equality holds and $\mathrm{E}[S_J] < \infty$.

5.5.3 G/G/1 queues

The intervals between new busy periods form a renewal process.

Conditional on $J = j$, let $X_{2,k} = X_{j+k}$, $V_{2,k-1} = V_{j+k-1}$ for $k \geqslant 1$, then $\{(X_{2,k}, V_{2,k-1}); k \geqslant 1\}$ is a sequence of i.i.d. pairs with the original distribution and statistically independent of J and $(X_i, V_{i-1}); 1 \leqslant i \leqslant J$.

Theorem 4

Consider a G/G/1 queue with finite expected interarrival time $\mathrm{E}[X]$ and finite expected service time $\mathrm{E}[V] < \mathrm{E}[X]$. Then the subsequence of arrivals that see an empty system forms a renewal process. The expected number of arrivals $\mathrm{E}[J]$ between arrivals to an empty system is finite and the expected time between successive busy periods (i.e., the expected inter-renewal time) is equal to $\mathrm{E}[J]\mathrm{E}[X]$.

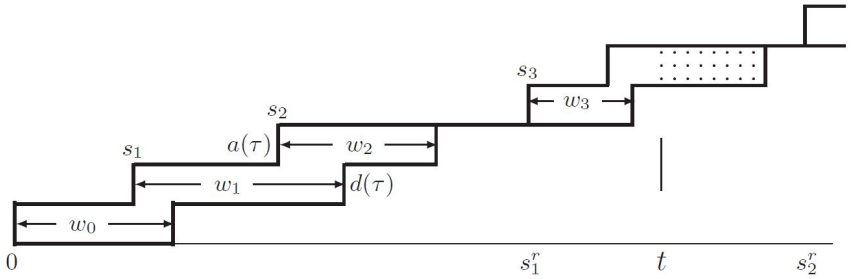
5.5.3 G/G/1 queues

Remark:

- The intervals between new busy periods form a renewal process. We then have one renewal process embedded in another. Call one the arrival process and the other the renewal process. The renewal process embodies both arrivals and services.
- Although successive pairs (X_i, Y_i) are assumed independent, it is not necessary for X_i and Y_i to be independent. This lack of independency does not occur for the G/G/1 (or G/G/m) queue.
- Nothing in the argument above for the G/G/1 queue made any use of the FCFS service discipline. One can use any service discipline for which the choice of which customer to serve at a given time t is based solely on the arrival and service times of customers in the system by time t.

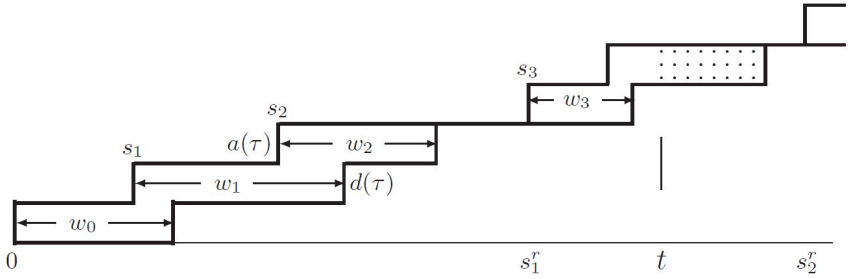
5.5.4 Little's Theorem

Consider a queueing system where the arrival process is a renewal process $A(t)$. Assume an arrival at time 0, thus $A(t) = N(t) + 1$.
The service process can be almost anything, but assume a G/G/1 queue to be specific, denoted by $D(t)$.
Assume the system empties out eventually WP1 and that it restarts on the next arrival.



We have seen that not only are the inter-renewal intervals $X_i^r = S_i^r - S_{i-1}^r$ i.i.d., but also the numbers of customer arrivals in the subsequent inter-renewal intervals are i.i.d..

5.5.4 Little's Theorem

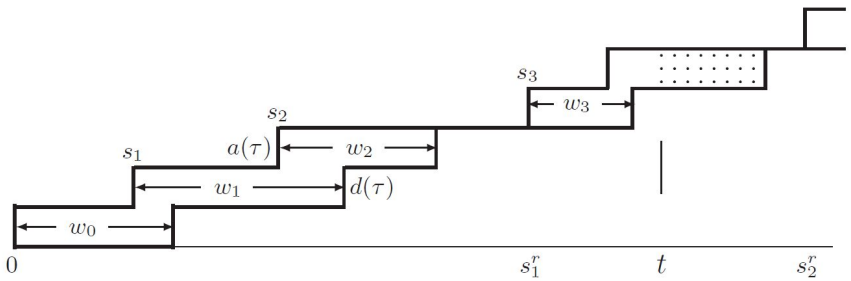


Let $L(\tau) = A(\tau) - D(\tau)$.

This depends on the departure process, but its value is a function of the inter-arrivals and service times within the current inter-renewal period. Thus it can be viewed as a generalized renewal-reward function on $N^r(t)$.

The total reward within an inter-renewal period L_n is then the integral of $L(\tau)$ over that period.

5.5.4 Little's Theorem



For the sample value in the figure, this integral is

$$\int_0^{s_1^r} L(\tau)\mathrm{d}\tau = w_0 + w_1 + w_2.$$

We can see that

$$\int_0^{S_1^r} L(\tau)\mathrm{d}\tau = \sum_{i=0}^{N(S_1^r)-1} W_i.$$

The same relationship exists in each inter-renewal interval, and in particular we define L_n for each $n \geqslant 1$.

$$L_n \triangleq \int_{S_{n-1}^r}^{S_n^r} L(\tau)\mathrm{d}\tau = \sum_{i=N(S_{n-1}^r)}^{N(S_n^r)-1} W_i.$$

The function $L(\tau)$ has the same behaviour as a renewal reward function, but it is slightly more general, being a function of more than the age of duration the renewal of the renewal counting process $\{N^r(t); t \geq 0\}$ at $t = \tau$.

However, the fact that $\{L_n; n \geq 1\}$ is an IID sequence lets us use the same methodology to treat $L(\tau)$ as was used earlier to treat renewal-reward functions.

5.5.4 Little's Theorem

Theorem 5 (Little's theorem)

For a FCFS G/G/1 queue in which the expected inter-renewal interval is finite, the limiting time-average number of customers in the system is equal, WP1, to a constant denoted as $\overline{L}$. The sample-path-average waiting time per customer is also equal, WP1, to a constant denoted as $\overline{W}$. Finally $\overline{L} = \lambda \overline{W}$ where λ is the customer arrival rate, i.e., the reciprocal of the expected interarrival time.

proof:

$$\sum_{n=1}^{N^r(t)} L_n \leqslant \int_{\tau=0}^{t} L(\tau)\mathrm{d}\tau \leqslant \sum_{i=0}^{N(t)} W_i \leqslant \sum_{n=1}^{N^r(t)+1} L_n.$$

$$\lim_{t\to\infty} \frac{\sum_{i=0}^{N(t)} W_i}{t} = \lim_{t\to\infty} \frac{\int_{\tau=0}^{t} L(\tau)\mathrm{d}\tau}{t} = \frac{\mathrm{E}[L_n]}{\mathrm{E}[X^r]} \quad \text{WP1}.$$

5.5.4 Little's Theorem

Thus, the limiting time average of $L(\tau)$ exists WP1 and is equal to

$$\overline{L} = \mathrm{E}[L_n]/\mathrm{E}[X^r].$$

On the other hand, we have

$$\lim_{t\to\infty} \frac{\sum_{i=0}^{N(t)} W_i}{t} = \lim_{t\to\infty} \frac{\sum_{i=0}^{N(t)} W_i}{N(t)} \lim_{t\to\infty} \frac{N(t)}{t}.$$

- The left term exists (and equals $\overline{L}$) WP1.
- The second limit on the right also exist WP1 by the strong law for renewal process. This limit is called the arrival rate $\lambda = 1/\overline{X}$.
- Thus the first limit on the right, which is the sample-path-average waiting time per customer, denoted $\overline{W}$, also exists WP1.

Thus, we have $\overline{L} = \lambda\overline{W}$.
(直观意义：系统人数等于到达速率 * 呆在系统的时间)。

5.5.4 Little's Theorem

Remark:

- In essence, Little's theorem is an accounting identity. Over a busy period and the following idle period (i.e., an inter-renewal period), $\int L(\tau)\mathrm{d}\tau = \sum_i W_i$. The difficult thing, mathematically, was showing that $\{N^r(t); t > 0\}$ is actually a renewal process and showing that the L_n are i.i.d., and this was where we needed to understand stopping rules.
- The question is not whether $\overline{L} = \lambda \overline{W}$, but whether these quantities exist as sensible time averages WP1 or as limiting ensemble averages.
- It is clear that FCFS order is not required for the argument. Thus the theorem generalizes to systems with multiple servers and arbitrary service disciplines in which customers do not follow FCFS order.

5.5.5 M/G/1 queues

- The arrival sequence $\{X_i; i \geqslant 1\}$ is Poisson with rate $\lambda = 1/\overline{X}$.
- Let $\{V_i; i \geqslant 0\}$ be the sequence of service requirements; these are i.i.d. and also independent of the arrivals.
- Assume that $\overline{X} > \overline{V} > 0$.
- the first arrival $J > 0$ to see an empty system is the smallest $n > 0$ for which $\sum_{i=1}^{n}(X_i - V_{i-1}) \geqslant 0$.

Theorem 6

For an M/G/1 queue with $\overline{V} < \overline{X} < \infty$, the expected number of arrivals $\mathrm{E}[J]$ and the expected time $\overline{X}\mathrm{E}[J]$ between successive initiations of busy periods is given by

$$\mathrm{E}[J] = \frac{\overline{X}}{\overline{X} - \overline{V}}, \qquad \overline{X}\mathrm{E}[J] = \frac{\overline{X}^2}{\overline{X} - \overline{V}}.$$

5.5.5 M/G/1 queues

Proof:

By the memorylessness of the Poisson process, the time from the completion of service of customer $J-1$ to the arrival of customer J is exponential with mean $\overline{X}$. What this means is that

$$\mathrm{E}\left[\sum_{i=1}^{J} X_i - \sum_{i=0}^{J-1} V_i\right] = \overline{X}.$$

Applying Wald's equality to $\{X_i - V_{i-1}; i \geqslant 1\}$, we see that

$$(\overline{X} - \overline{V})\mathrm{E}[J] = \overline{X}.$$

Thus,

$$\mathrm{E}[J] = \frac{\overline{X}}{\overline{X} - \overline{V}}, \qquad \overline{X}\mathrm{E}[J] = \frac{\overline{X}^2}{\overline{X} - \overline{V}}.$$

The Pollaczek-Khinchin formula for M/G/1 queues

- Assume that the service time V has a finite second moment $\mathrm{E}[V^2] < \infty$.
- Let $L^q(t)$ be the number of customers in queue and let $R(t)$ be the residual life of the customer in service.
- Let $U(t)$ be the waiting time in queue that would be experienced by a customer arriving at time t, assuming FCFS service.

Thus,

$$U(t) = \sum_{i=0}^{L^q(t)-1} V_{N(t)-i} + R(t).$$

Note that if the queue contains $L^q(t) \geqslant 1$ customers at time t, then customer number $N(t)$ is at the back of the queue and customer number $N(t) - L^q(t) + 1$ is at the front. 截至 t, 共有 $N(t)$ 客户到达，而队列长度是 $L^q(t)$, 完成服务或正在接受服务的顾客数目是 $N(t) - L^q(t)$, 因此此时排在队首的是 $N(t) - L^q(t) + 1$，而队尾的是 $N(t)$。

The Pollaczek-Khinchin formula for M/G/1 queues

Since the service times and inter-arrival times of these completed services are independent of the service times of the customers in the queue, and thus, given $L^q(t) = \ell$,

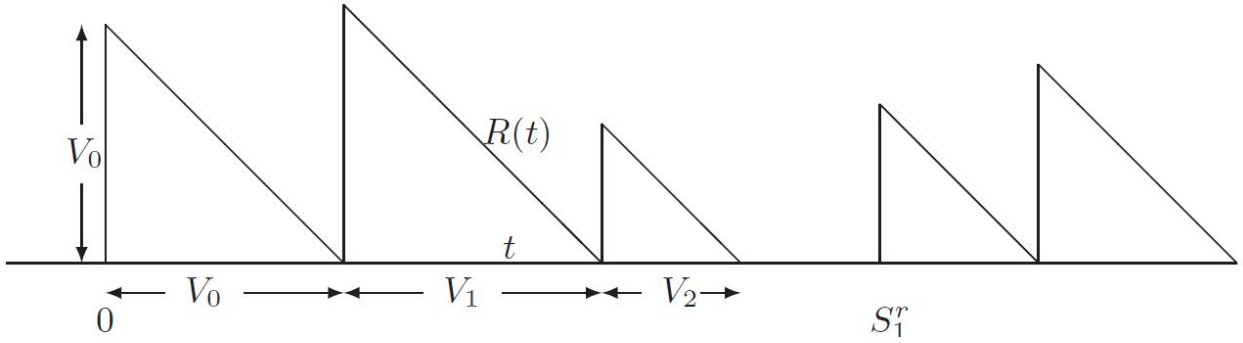
$$\mathrm{E}\left[\sum_{i=0}^{L^q(t)-1} V_{N(t)-i} \middle| L^q(t) = \ell\right] = \ell \overline{V}.$$

Taking the expectation over $L^q(t)$,

$$\mathrm{E}[U(t)] = \mathrm{E}[L^q(t)]\overline{V} + \mathrm{E}[R(t)].$$

We first calculate the time average of $R(t)$ by viewing it as a reward function.

The Pollaczek-Khinchin formula for M/G/1 queues



Note that $S^r_{N^r(t)}$ is not only a renewal epoch for the renewal process, but also an arrival epoch for the arrival process; in particular, it is the $N(S^r_{N^r(t)})$-th arrival epoch, and the $N(S^r_{N^r(t)})-1$ earlier arrivals are the customers that have received service up to time $S^r_{N^r(t)}$.

$$\int_0^{S^r_{N^r(t)}} R(\tau)\mathrm{d}\tau = \sum_{i=0}^{N(S^r_{N^r(t)})-1} \frac{V_i^2}{2} \leqslant \sum_{i=0}^{N(t)-1} \frac{V_i^2}{2} \leqslant \int_0^{S^r_{N^r(t)+1}} R(\tau)\mathrm{d}\tau$$

The Pollaczek-Khinchin formula for M/G/1 queues

From going through virtually the same argument several times, we can prove that

$$\lim_{t\to\infty} \frac{\int_0^t R(\tau)\mathrm{d}\tau}{t} = \lim_{t\to\infty} \frac{\sum_{i=0}^{N(t)-1} V_i^2}{2N(t)} \frac{N(t)}{t} = \frac{\lambda \mathrm{E}[V^2]}{2} \quad \text{WP1.}$$

As will be seen in Section 5.7.5, the time average above can be replaced with a limiting ensemble average so that

$$\lim_{t\to\infty} \mathrm{E}[R(t)] = \frac{\lambda \mathrm{E}[V^2]}{2}.$$

Similarly, there is a limiting ensemble-average form of Little' s theorem which shows that

$$\lim_{t\to\infty} \mathrm{E}[L^q(t)] = \lambda \overline{W^q}.$$

The Pollaczek-Khinchin formula for M/G/1 queues

Thus,

$$\lim_{t\to\infty} \mathrm{E}[U(t)] = \lambda \mathrm{E}[V^2]\overline{W^q} + \frac{\lambda \mathrm{E}[V^2]}{2}.$$

注意:

- $\mathrm{E}[U(t)]$ is the expected unfinished work at time t; this is the queueing delay that a customer would incur by arriving at t.
- $\overline{W^q}$ is the sample-path-average expected queueing delay.

For Poisson arrivals, the probability of an arrival in $(t, t+\delta]$ is independent of all earlier arrivals and service times, so it is independent of $U(t)$. Thus $\lim_{t\to\infty} \mathrm{E}[U(t)]$ must be equal to $\overline{W^q}$.
Thus, we obtain the Pollaczek-Khinchin formula:

$$\overline{W^q} = \frac{\lambda \mathrm{E}[V^2]}{2(1-\lambda \mathrm{E}[V])}.$$

The "slow truck effect"

If $\mathrm{E}[V^2] = \infty$, the limiting expected queueing delay is infinite even when the expected service time is less than the expected interarrival interval.

Why does $\overline{W^q}$ go up with $\mathrm{E}[V^2]$?

Have you ever noticed, when entering a line for service that the customer being served often takes much longer than anyone else?

Note that while a particularly long service is taking place, numerous arrivals are coming into the system, and all are being delayed by this single long service. In other words, the number of new customers held up by a long service is proportional to the length of the service, and the amount each of them is held up is also proportional to the length of the service.
非泊松到达的例子： V 在 $(1-\epsilon, 1+\epsilon)$ 均匀分布，到达在 $(2-\epsilon, 2+\epsilon)$ 均匀分布。对于 $\epsilon < 1/2$, 无队列，但。。。

Thank you for your attention!